**TASK 4: SENTIMENT ANALYSIS**

**Course:** Data Analysis
**Project Title:** Sentiment Analysis of Book Reviews
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**Tools:** Python, Pandas, VADER Sentiment Analyzer, Matplotlib, Visual Studio Code
**Dataset:** Sample Book Reviews Dataset
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# 1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to identify the emotional tone expressed in written text. It classifies text into categories such as Positive, Negative, and Neutral.

In this task, sentiment analysis was performed on a sample dataset of book reviews using Python and the VADER Sentiment Analyzer. The purpose was to classify reviews, examine sentiment patterns, visualize the results, and understand how sentiment analysis can support marketing and product development decisions.

# 2. Objectives

The main objectives of this task were:

- To analyze text data and classify it as positive, negative, or neutral.

- To apply NLP techniques and a lexicon-based sentiment analysis method.
- To calculate sentiment scores for individual book reviews.
- To visualize the distribution of sentiment categories.
- To understand how sentiment patterns can help summarize customer opinions.
- To explore applications in marketing, product improvement, and social insights.
- To identify the limitations of automated sentiment analysis.

## 3. Tools and Technologies

**Python:** Used to implement the sentiment analysis program.

**Pandas:** Used to load, clean, process, and organize the review dataset.

**VADER Sentiment Analyzer:** A lexicon-based NLP tool used to calculate sentiment scores and classify English text.

**Matplotlib:** Used to create charts for presenting sentiment results.

**Visual Studio Code:** Used to write and execute the Python code.

**CSV:** Used to store the input reviews and the resulting sentiment classifications.

## 4. Dataset Description

A sample dataset named sentiment_reviews.csv was created for this task. It contains 30 example book reviews, each identified by a unique review ID.

The dataset includes positive, negative, and neutral statements. These reviews provide sample text for demonstrating sentiment classification.

The dataset is intended for educational demonstration. It does not contain verified customer feedback collected from Amazon or social media, and its results cannot be considered representative of actual public opinion.

```
PS C:\Users\PC\Desktop\Web_Scraping_Task> python sentiment_analysis.py
Negative    33.33
Neutral     33.33
Name: count, dtype: float64

Files saved successfully:
- sentiment_analysis_results.csv
- sentiment_distribution.png
- sentiment_percentages.png
- sentiment_summary.txt
```

## 5. Methodology

The following steps were performed:

1. The review dataset was loaded using Pandas.
2. Missing and empty review text was checked and handled.
3. The VADER Sentiment Analyzer was initialized.
4. Positive, negative, neutral, and compound sentiment scores were calculated for each review.
5. Each review was classified as Positive, Negative, or Neutral using the compound score.
6. Sentiment counts and percentages were calculated.
7. The results were saved in a CSV file.
8. Charts were created to visualize the distribution of sentiment categories.

## 6. Natural Language Processing and Lexicon-Based Analysis

Natural Language Processing enables computers to process and analyze human language. In this task, VADER was used as a lexicon-based sentiment analysis tool.

A sentiment lexicon contains words associated with positive or negative sentiment. VADER evaluates these words along with features such as punctuation, capitalization, and certain language patterns to estimate the sentiment of a text.

The compound score ranges from -1 to +1. The following thresholds were used:

- Positive: Compound score greater than or equal to 0.05.
- Negative: Compound score less than or equal to -0.05.
- Neutral: Compound score between these thresholds.

For example, a review stating that a book is “amazing” may receive a positive classification, while a review describing a book as “terrible” may receive a negative classification. A factual statement about the number of pages may be classified as neutral.

## 7. Sentiment Classification Results

The program assigned a sentiment label to each review based on its compound score. The output includes the review ID, original review text, positive score, negative score, neutral score, compound score, and final sentiment label.

The results were saved in sentiment_analysis_results.csv for further examination.

The number and percentage of reviews in each sentiment category were calculated to summarize the overall distribution within the sample dataset.

## 8. Sentiment Distribution Visualization

A bar chart was created to compare the number of Positive, Negative, and Neutral reviews. It provides a clear comparison of the frequency of each sentiment category.

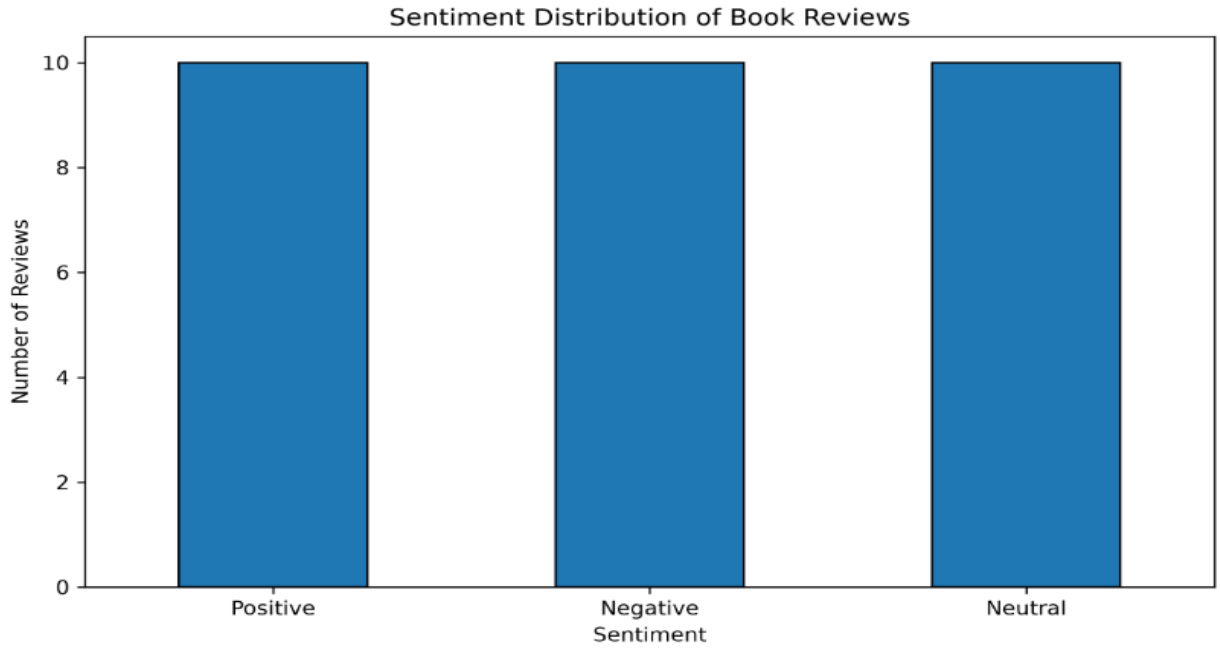


**Figure 1: Distribution of Sentiment Categories**

A pie chart was also created to display the percentage of reviews belonging to each sentiment category.

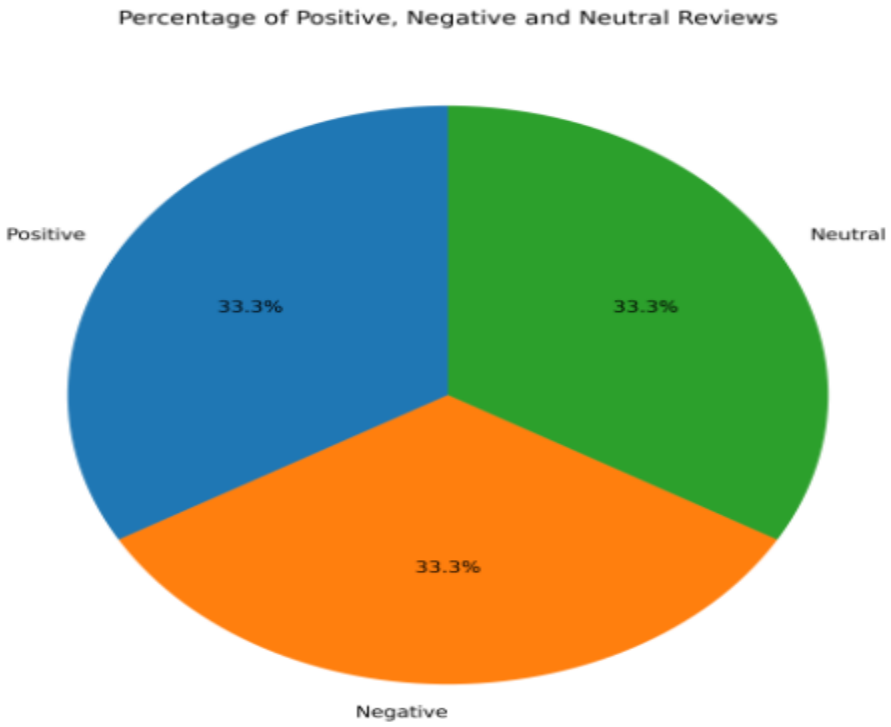



**Figure 2: Percentage Distribution of Sentiments**

Together, these charts help communicate the sentiment distribution in a simple and understandable format.

**9. Understanding Public Opinion and Trends**

Sentiment analysis can help organizations summarize opinions expressed in customer reviews, social media posts, and other text sources.

Positive sentiment may indicate satisfaction or appreciation, while negative sentiment may reveal dissatisfaction or possible areas for improvement. Neutral sentiment may reflect factual or descriptive statements.

In this task, the analysis demonstrates how sentiment patterns can be summarized. However, because the dataset contains sample reviews rather than real customer or social media data, it does not establish actual public opinion or real-world trends.

**10. Applications in Marketing and Product Development**

Sentiment analysis can support business decisions in several ways:

**Marketing:** Organizations can analyze genuine customer feedback to understand how people respond to a product or campaign.

**Product Development:** Repeated negative comments may help identify features that require investigation or improvement.

**Customer Feedback:** Positive and negative reviews can be summarized to identify common areas of satisfaction and dissatisfaction.

**Social Insights:** When applied to representative social media data, sentiment analysis can help monitor changes in public reactions over time.

These applications should be supported by reliable data and human review because automated sentiment classification may make mistakes.

### 11. Limitations

Sentiment analysis has several limitations:

- Sarcasm and irony may be classified incorrectly.
- The meaning of words can change depending on context.
- Mixed positive and negative opinions may be difficult to classify.
- VADER estimates overall sentiment polarity and does not reliably identify every specific emotion, such as fear, sadness, or joy.
- The sample dataset is small and does not represent all readers or customers.
- Automated predictions should not be treated as perfectly accurate.

### 12. Results and Findings

The task successfully demonstrated sentiment classification using a lexicon-based NLP tool. The program calculated sentiment scores, assigned labels, saved the results in a CSV file, and generated two visualizations.

The actual sentiment counts and percentages should be taken from the program's terminal output and used to describe which category was most common in the sample dataset.

The results demonstrate how text data can be converted into structured information for easier interpretation.

### 13. Conclusion

This task demonstrated the practical application of Natural Language Processing in sentiment analysis. Python, Pandas, VADER, and Matplotlib were used to process sample book reviews, classify their sentiment, and present the results through visualizations.

The exercise showed how sentiment analysis can help organize and summarize textual feedback. It also highlighted the importance of data quality, context, and appropriate interpretation when using automated tools to support marketing, product development, and social insight activities.